📘 Day 17: Endpoint Security (EDR & XDR)

🎯 Objective

Understand how endpoint security tools detect, analyze, and respond to cyber threats on devices.

🔐 1. Endpoint Security

Endpoint security focuses on protecting individual devices such as:

- Laptops
- Desktops
- Servers
- Mobile devices

👉 These endpoints are common entry points for attackers, so securing them is critical.

💻 2. EDR (Endpoint Detection and Response)

🔍 What is EDR?

EDR is a security solution that continuously monitors endpoint activities to detect suspicious behavior.

⚙️ Key Functions:

- **Process Monitoring** → Tracks running applications
- **Threat Detection** → Identifies malicious activity
- **Response Actions**:
  - Kill malicious process
  - Isolate infected system

- Alert SOC team

💡 Example:

- Suspicious file: malware.exe
- Behavior: connects to unknown external IP
- Action: process terminated + system isolated

🌐 3. XDR (Extended Detection and Response)

🔍 What is XDR?

XDR extends EDR by collecting and correlating data from multiple sources:

- Endpoints
- Network traffic
- Emails
- Servers

🚀 Benefits:

- Better visibility across systems
- Faster detection of complex attacks
- Reduced false positives
- Centralized security analysis

⚖️ 4. EDR vs XDR

Feature	EDR	XDR
Scope	Endpoints only	Multiple sources (endpoint + more)
Data Sources	Device-level	Endpoint, network, email
Detection	Individual device threats	Advanced multi-vector attacks
Visibility	Limited	Broad & centralized

🧠 5. Detection Logic (Basic Rule)

Example logic used in SOC tools:

IF unknown process
AND external connection
THEN trigger alert

👉 This is how SIEM/EDR systems identify suspicious behavior patterns.

🔄 6. SOC Workflow

👮 Incident Handling Flow:

1. **L1 Analyst**
    a. Detect alerts
    b. Perform initial triage
2. **L2 Analyst**
    a. Investigate deeper
    b. Confirm if it's a real threat
3. **L2 Response**
    a. Take action (block, isolate, remove)
4. **L3 Analyst**
    a. Improve detection rules
    b. Threat hunting & tuning

⚔️ 7. Attack Scenario Example

- User downloads a malicious file
- File executes (malware.exe)
- Connects to attacker server
- Starts data exfiltration

👉 **Detection:**

- EDR flags unusual process + connection

👉 **Response:**

- Kill process
- Isolate system
- Block IP
- Alert SOC team

🛡️ 8. Recommended Response Actions

- Isolate affected endpoint
- Terminate malicious processes
- Block suspicious IP/domain
- Scan system for malware
- Update detection rules

📝 Assignment Answers

1. EDR vs XDR

- EDR focuses only on endpoint-level monitoring and response
- XDR provides broader security by integrating multiple data sources

2. Attack Scenario

- Malware execution + external communication + data theft

3. Suggested Response

- Kill process, isolate device, block IP, investigate logs